

AI • 104

Support & Operations Professionals.

Building AI literacy and embracing the new professional paradigm.

THE PARADIGM SHIFT

Two eras, two **operating models**.



The Software Era

Humans write every rule. Logic is explicit, predictable, and brittle outside its envelope.

OPS EXAMPLE

Static FAQ pages and standard operating procedures.



The AI Era

Humans provide examples; machines learn the rules. Behaviour adapts to context the team didn't pre-anticipate.

OPS EXAMPLE

Dynamic assistants that understand customer frustration and adapt.

WHY AI IS NO LONGER OPTIONAL IN OPERATIONS

Three forces driving the **non-negotiable**.



01 • SCALE

24/7 demand exceeds human throughput.

Human teams cannot keep up with the always-on, global demand of modern digital business. AI acts as the first responder.



02 • DATA OVERLOAD

More logs than any team can read.

We generate more logs, tickets, and emails than any human can read. We need AI to triage this massive influx.



03 • CAREER RESILIENCE

From task-doer to orchestrator.

Understanding AI elevates you from a task-doer to an orchestrator, significantly increasing your professional value.

A WORKING DEFINITION

AI is the field of CS that builds systems capable of **human-like tasks**.

Three capabilities define the boundary:

CAPABILITY 01
Learning

CAPABILITY 02
Reasoning

CAPABILITY 03
Self-correction



Standard Software

Follows a rigid, set formula for predictable inputs. Behaviour is deterministic and falls over the moment reality is messy.



Artificial Intelligence

Interprets messy, real-world data — like a blurry photo or a sarcastic email — and produces a useful response.

THE CAPABILITY MATRIX

Weak vs. Strong AI – and where **we actually live.**

Feature	Weak / Narrow AI	Strong / General AI
DEFINITION	Designed and trained for a specific task.	Full cognitive abilities of a human.
AWARENESS	No consciousness or genuine understanding.	Self-aware; can reason, plan, and feel.
EXAMPLES	Siri, Spam Filters, ChatGPT, Support Bots.	JARVIS. HAL 9000.
CURRENT STATUS	Everywhere. This is what we use today.	Theoretical. It does not exist yet.

KEY MINDSET

Direct the technology before **it directs you.**

AI isn't replacing support & ops professionals — it's eliminating the boring parts so humans can focus on the complex, creative, and human elements that machines genuinely cannot replicate.

✓ What AI does well

- Process large volumes of routine requests without fatigue.
- Surface patterns humans would miss across thousands of data points.
- Draft consistent first responses at scale.
- Classify and route with high accuracy when trained correctly.
- Summarize long threads and incident logs instantly.

⚠ What AI still can't do

- Exercise genuine empathy in complex emotional situations.
- Make judgment calls that require full business context.
- Handle novel edge cases it hasn't been trained on reliably.
- Take accountability for outcomes.
- Replace the human relationship in high-stakes escalations.

TWO MODES OF OPERATIONAL AI

The Forecaster & the Creator.



Predictive AI – the Forecaster

- Uses statistical algorithms to find **if-this-then-that** correlations in massive datasets.
- Doesn't create – it categorizes or calculates probability.

OPS EXAMPLE

Analyzes ticket sentiment to predict if a customer is at high risk of canceling based on past frustration.



Generative AI – the Creator

- Learns the distribution of data from billions of examples.
- Predicts the next likely piece of data to build original output.

OPS EXAMPLE

A pro provides bullet points; AI generates a full, formatted SOP in seconds.

GENERATIVE AI UNDERSTANDS CONTEXT, BUT REQUIRES EXPLICIT INSTRUCTION

System types & the context-window funnel.

SYSTEM TYPE 01

Large Language Models (Gen AI)

Understand & generate text, adapt to novel situations — but make confident mistakes.

SYSTEM TYPE 02

Classifiers (Classic AI)

Route tickets based on historical rules. High precision on clean data.

The context-window funnel

All information fed into the AI — instructions, history, documents — must fit within the context window. Prompt constraints shape the refined output.

INSTRUCTIONS • SYSTEM PROMPT

CONVERSATION HISTORY

REFERENCE DOCUMENTS

REFINED OUTPUT

FOUR CORE CONCEPTS

The vocabulary of **working with LLMs**.

CONCEPT 01

Prompt

The instruction you give. Quality determines output.

CONCEPT 02

Context Window

How much information the AI can hold in mind at once.

CONCEPT 03

Hallucination

Plausible-sounding but factually incorrect generated information.

CONCEPT 04

Temperature / Confidence

Controls creativity vs. consistency. Lower = consistent; higher = varied.

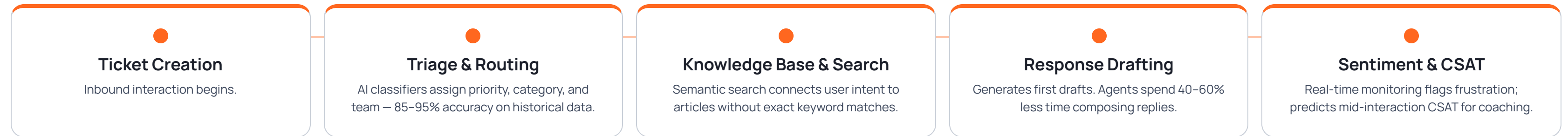
PRO TIP

Think of an LLM like a brilliant intern who has read everything ever written — but has never actually done the work.

They need supervision and explicit context. Always.

SUPPORT LIFECYCLE

AI intersects every stage of modern support.



Agent remains the final sender — AI surfaces signal, prioritizes, and drafts; humans approve.

THE CHATBOT EFFICACY MATRIX

Knowing **when to automate.**

✓ Ideal conditions – chatbots work well

- High-volume, repetitive queries (password resets, order status).
- 24/7 coverage outside business hours.
- First-level triage before human handoff.
- Guided troubleshooting with clear decision trees.

✗ Failure modes – chatbots fail

- Complex, multi-issue problems requiring judgment.
- Emotionally distressed or frustrated customers.
- Novel situations not in training data.
- Nuanced billing or legal disputes.

OPS LIFECYCLE

Catch issues earlier. **Resolve them faster.**



REAL SCENARIO

02:31

Database config change deployed.

02:47

Payment service returns elevated error rates.

02:48
(+90s)

AI monitoring detects the anomaly.
Correlates with the 02:31 deploy, pages on-call with full context, and suggests rollback. **Zero customer complaints filed.**

AI MANIFESTED

The operations toolkit.



Patterns in Data

Concept: finding the needle in the haystack.

Ops use: cybersecurity login anomaly detection.



Natural Language Processing

Concept: understanding human-speak.

Ops use: auto-tagging and routing support tickets.



Internet of Things

Concept: connecting physical objects to the internet.

Ops use: smart factories & predictive maintenance.



Generative Systems

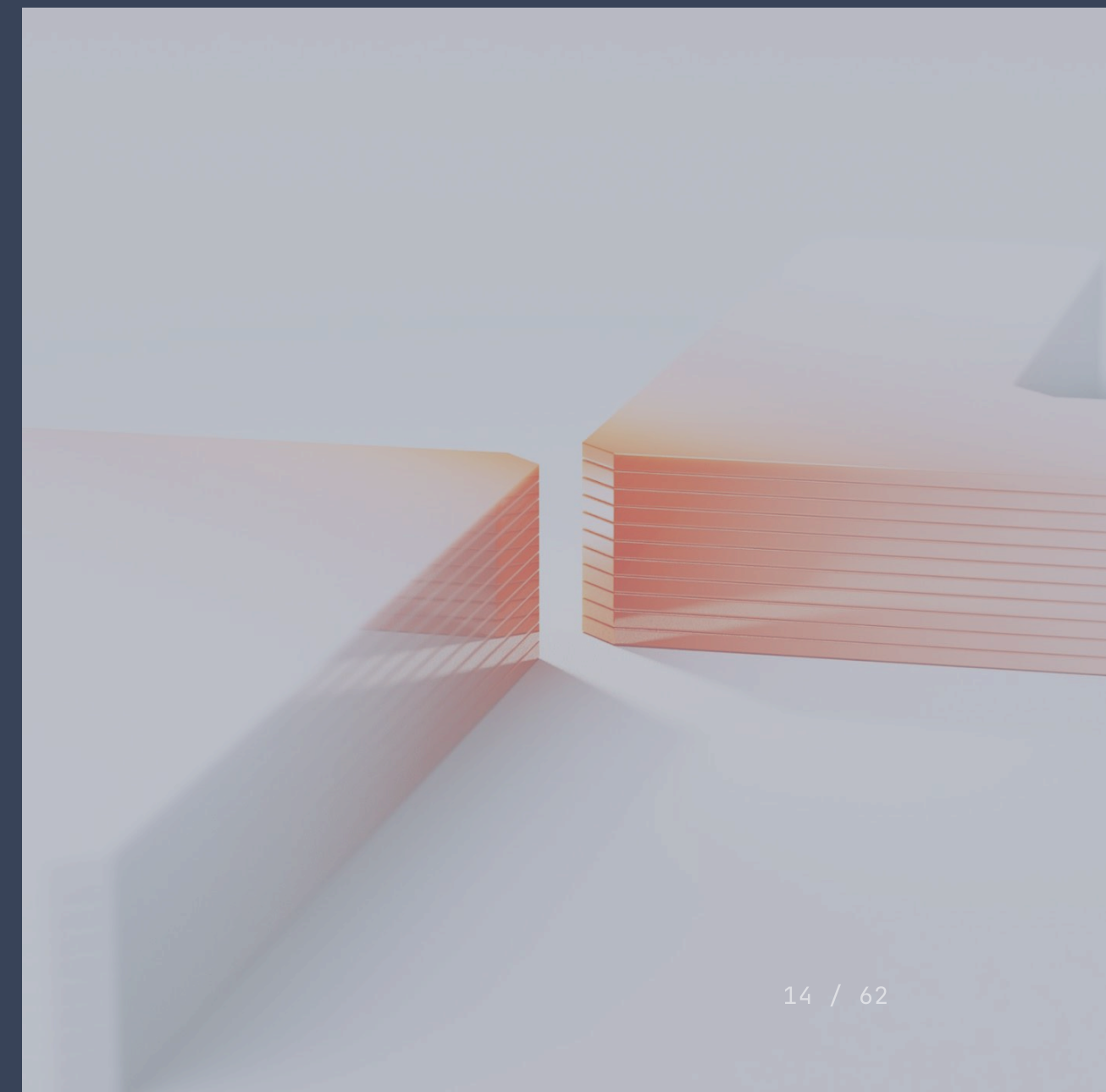
Concept: foundational large language models.

Ops use: synthetic data for safe system testing.

PART 02 • MODULE

Artificial Intelligence and Machine Learning.

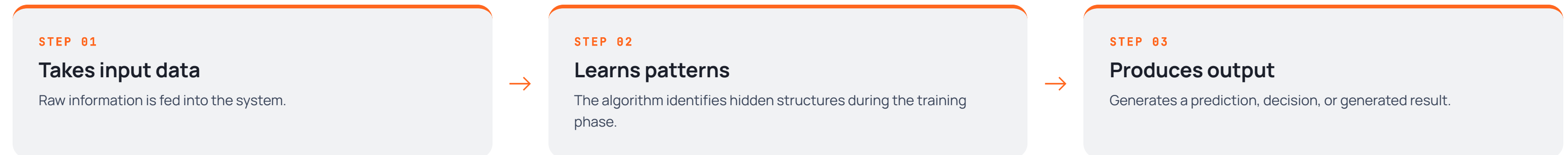
Foundations, vocabulary, and the jobs each model class is actually good at.



WHAT AN AI MODEL IS

Math that learns patterns — **not rules** **someone wrote.**

AI models are mathematical algorithms that learn patterns from data to make decisions, without being explicitly programmed.



THREE CORE LEARNING STYLES

How models **learn**.

01

Supervised Learning

Data: labeled (input + correct output).

Goal: predict correct answers.

Email spam detection (Gmail).

02

Unsupervised Learning

Data: unlabeled.

Goal: find hidden patterns or groups.

Customer segmentation by buying behavior (Amazon).

03

Reinforcement Learning

Data: trial & error (rewards / penalties).

Goal: maximize reward over time.

Game-playing AI like AlphaGo.

PRIMARY TASK CATEGORIES

Four output shapes — categorize, predict, group, recommend.

CLASSIFICATION

Categories & labels

Output: discrete categories.

Fraud detection in PayPal — transaction → fraud / not fraud.

REGRESSION

Continuous values

Output: a real-valued number.

Zillow house-price prediction from features.

CLUSTERING

Groups of similar data

Output: unsupervised groupings.

Spotify clusters songs by listening patterns.

RECOMMENDATION

Personalized suggestions

Output: ranked items.

Netflix movie suggestions based on behavior.

ALGORITHM GALLERY – CLASSIFICATION MODELS

Four classifiers, four **jobs**.

DECISION TREE

If-then rules

Classify using a series of if-then rules. Easy to explain to non-technical stakeholders.

rules-based · medical diagnosis · loan approval

RANDOM FOREST

Many trees, voted

More accurate than a single tree, less prone to overfitting.

tabular data · fraud detection · churn

KNN

Nearest neighbours

Classify by looking at the K most similar past examples. No training phase.

similarity-based · small dataset

LOGISTIC REGRESSION

Yes / No decisions

Classify into 2 categories. Despite the name, it's a classifier.

spam-or-not · churn-or-not · binary outcomes

ALGORITHM GALLERY — ADVANCED MODELS

From a straight line to **deep nets**.

LINEAR REGRESSION

Predict a number

When you expect a straight-line relationship.

predict price · forecast sales · how much / many

K-MEANS

Group unlabelled data

Find K natural groups in your data.

group customers · segment users · find patterns

NEURAL NET / DEEP LEARNING

Complex patterns

Images, text, audio. Needs lots of compute.

image recognition · speech · text generation

POLYNOMIAL / MULTIPLE

Many factors, curves

Predict a number using multiple features when the relationship curves.

house price with many features · curved trend

THE COMPLEXITY EVOLUTION

From decision trees to **generative systems.**

TIER 01

Traditional ML

Simpler, foundational models — Linear Regression, Decision Trees.

Predicting sales trends in retail.



TIER 02

Deep Learning

Built on multi-layered neural networks. Higher compute.

Face recognition technology (Facebook).



TIER 03

Generative AI

Systems capable of synthesizing entirely novel outputs.

Chatbots like ChatGPT generating answers and code.

THE AI TAXONOMY

Master cheat sheet.

Classification type	Core focus	Real-world example
SUPERVISED	Labeled data	Spam detection
UNSUPERVISED	Pattern discovery	Customer groups
REINFORCEMENT	Reward-based learning	Game AI
CLASSIFICATION	Categories	Fraud detection
REGRESSION	Numbers	Price prediction
DEEP LEARNING	Complex patterns	Face recognition
GENERATIVE AI	Content creation	Chatbots

MACHINE LEARNING – THE ENGINE INSIDE THE AI

Artificial Neural Networks **mimic neurons.**

They train themselves without fixed instructions – adjusting their internal weights as data flows through.



ON MODEL TRAINING

The goal of an AI model is not to memorize the past — but to **learn the patterns** necessary to predict the future.

The ultimate challenge of model training is balancing how well a system learns without simply memorizing data.

THE TWO FAILURE MODES OF TRAINING

Underfitting & overfitting.



Underfitting – the simple failure

The flaw: model is too simple; fails to learn the pattern.

Performance: fails on training data × | fails on new data ×

Root causes: not enough training time, missing critical features.



Overfitting – the complex failure

The flaw: model learns too much, including pure noise.

Performance: perfect on training data ✓ | fails on new data ×

Root causes: model is too complex, dataset is too small.

REAL-WORLD IMPACT — PREDICTING HOUSE PRICES

Three fits, **three outcomes.**



UNDERFIT

Looks at rooms only

Ignores location, size, and amenities. Predictions fail everywhere.



OVERFIT

Obsesses over noise

Memorizes a specific crack in one training house's driveway.
Cannot predict prices in a new neighborhood.



PERFECT FIT

True patterns

Identifies square footage, zip code, and market trends.
Generalizes cleanly.

INSIDE THE NEURAL NET · PART 01

The neuronal core.

01 · WEIGHTS

The dials of importance

Numerical values that determine how strongly a particular input influences the neuron's output.

02 · BIAS

The baseline shift

An extra parameter that shifts predictions, allowing the model to fit data flexibly — not forced through zero.

03 · ACTIVATION

The decision gate

Decides if a neuron should fire. Introduces non-linearity so the network can learn complex patterns.

INSIDE THE NEURAL NET · PART 02

The correction loop.

01 · LOSS FUNCTION

The judge

Measures exactly how wrong the model's prediction is compared to reality. The sole goal of training is to minimize this number.

02 · BACKPROPAGATION

The messenger

Takes the error from the loss function and propagates it backward through the network to update Weights and Bias.

03 · GRADIENT DESCENT

The navigator

The optimization technique that physically moves the weights in the mathematical direction of lowest error.

INSIDE THE NEURAL NET · PART 03

The **hyperparameters**.



LEARNING RATE

The step size

Controls how drastically weights are updated. **Too high** overshoots the solution. **Too low** takes forever.



EPOCH

The repetitions

One complete pass through the entire dataset. Multiple passes improve learning, but too many trigger overfitting.



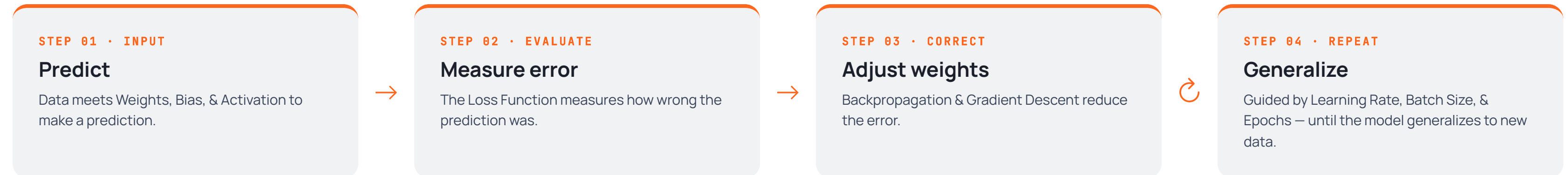
BATCH SIZE

The chunks

Number of samples processed at once. **Small** is fast but chaotic. **Large** is stable but slow.

SYNTHESIS

The model-training lifecycle.



FILTERING THE NOISE

Activation bias — the manager analogy.

Setting a filter

"Don't bother me with minor complaints; only come to me if the total loss is over \$500."

A neuron only fires if the total signal crosses a mathematical threshold — exactly the same principle.

The application

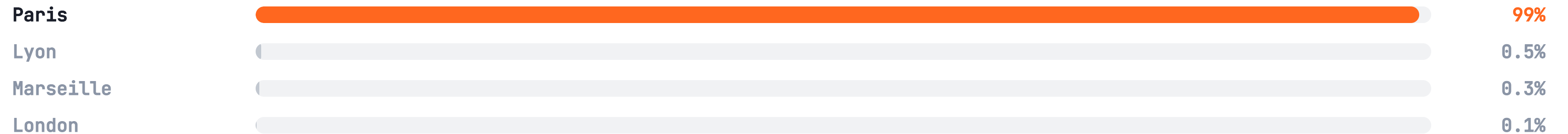
The **\$500** is the bias. It filters out the noise so the AI system stays focused entirely on what matters.

Without bias, every signal would shout equally loudly — and nothing would actually get learned.

THE PROBABILITY ENGINE

How LLMs **actually** work.

Prompt: "The capital of France is..."



KEY TAKEAWAY

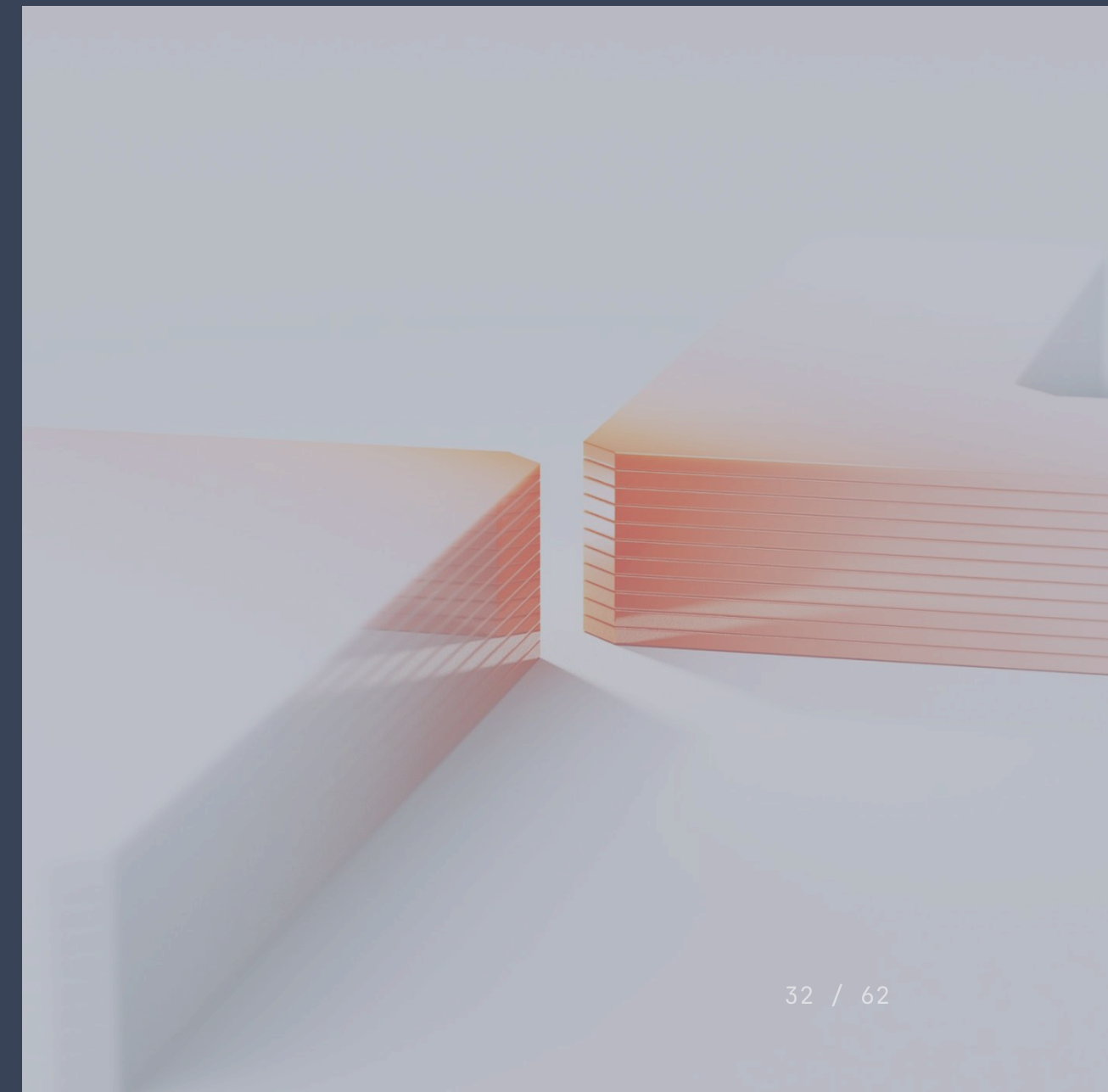
LLMs operate through pattern recognition and probability — not factual reasoning or conscious thought.

The model selects the next word based on statistics, not sentient thought.

PART 03 • MODULE

Prompt Engineering.

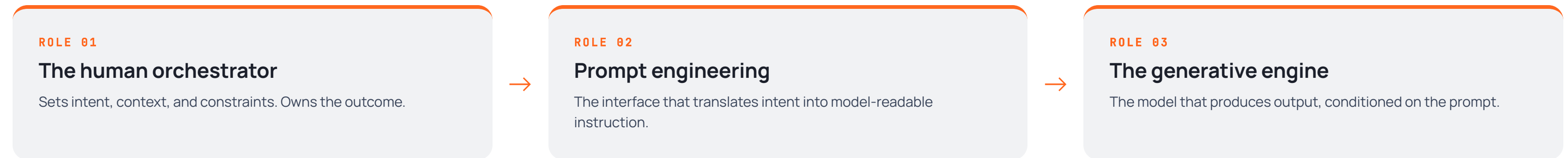
Design the question and the model becomes a colleague — not a coin flip.



PROMPT ENGINEERING – THE STEERING WHEEL

The human **orchestrator**.

We're now operating in a world where technology generates new data – text, code, images. Prompt engineering is how you steer it.



FEW-SHOT LEARNING IN ACTION • CLAUDE AI

Standardizing messy incident reports

The prompt: "Turn raw notes into formal reports. Follow this style exactly:"

RAW DATA

Server went down at 2pm...

FORMAL OUTPUT

[INCIDENT-001] | Status: Resolved | Duration: 60m...

BEYOND TEXT — VISUAL & VOICE OPERATIONS

Multi-modal prompting in **the field.**



Text-to-Image (T2I) prototyping

The shift: from hours searching stock art or waiting for designers to creating UI mockups in seconds.

Application: generating instant visual prototypes — Figma AI layouts or DALL-E server-rack visualizations — directly from plain English descriptions.



Multimodal voice — eyes-on assistant

User: "Status light on primary PSU is flashing green, secondary is solid amber on an R740. Hot-swap or firmware sync?"

AI: "Solid amber indicates predictive failure or power mismatch. Prep a replacement unit for a hot-swap immediately."

THE ANATOMY OF A PRODUCTION-GRADE PROMPT

Five parts. No exceptions.

01 • ROLE

Who the AI is

02 • CONTEXT

Background & history

03 • TASK

Explicit output wanted

04 • FORMAT

Output structure

05 • CONSTRAINTS

Guardrails & tone

WEAK PROMPT

"Write a reply to this angry customer."

STRONG PROMPT

[ROLE]: You are a senior support agent at a cloud SaaS company.

[TASK]: Draft a response to the following ticket...

[CONTEXT]: ...from a customer who has been waiting 3 days for a refund.

[CONSTRAINTS]: Tone: empathetic and apologetic. Acknowledge delay, explain timeline (5-7 days). Do not promise anything beyond policy.

[FORMAT]: Keep it under 150 words.

DIVING INTO CHATGPT-5

Your logic engine.

ChatGPT-5 is the most advanced model for multi-step reasoning and complex problem-solving.

KEY FEATURE

Deep Thinking mode

The model pauses and 'reasons' through a problem before answering. Ideal for multi-step analysis and debugging.

OPS USE CASE

Complex debugging

Microservices logs, sophisticated scripts, and runbooks — paste-and-reason flows.

POWER USER TIP

272k token window

Feed it an entire codebase or a 200-page technical manual in one go. Ideal for large-scale document analysis.

DIVING INTO CHATGPT-5 · EXAMPLE A

Use ChatGPT-5 as your **Senior Architect**.

It excels at complex reasoning and connecting disparate pieces of information.

EXAMPLE A · SCRIPT DEBUGGING

PROMPT

"You are a Senior Systems Engineer. I have a Python script that is supposed to automate user offboarding in Azure, but it's failing at the 'Revoke License' step with a 403 error. Here is the code: [Paste Code]. Analyze the logic, identify the most likely permission mismatch, and provide the corrected snippet."

 THINKING · GPT-5 REASONING ENGINE

- User is performing Azure AD (Entra ID) offboarding.
- Error is 403 Forbidden during the license revocation step.
- Script uses MS Graph API via the **msal** library.
- Common pitfall: 403 usually implies the Service **Directory.ReadWrite.All** or **User.ReadWrite.All** – but for licenses specifically, **Organization.Read.All** is often missed. Principal lacks

DIVING INTO CHATGPT-5 · EXAMPLE A

Analysis: the 403 Forbidden error.

Based on the code provided, the 403 error isn't a syntax bug — it's a **scope mismatch**.

Root cause

Your Service Principal is likely missing the **Organization.Read.All** permission. While **User.ReadWrite.All** lets you edit the user, the license-assignment endpoint needs to *read* the organization's SKU details to verify which licenses are being removed.

Required Microsoft Graph permissions

01 · APPLICATION

User.ReadWrite.All

Edit user objects in the directory.

02 · APPLICATION

Directory.ReadWrite.All

Read & write directory data.

03 · MISSING PIECE

Organization.Read.All

Read tenant SKU & license data.

DIVING INTO CHATGPT-5 · EXAMPLE A

Corrected code — logic update.

Ensure your header includes the correct scope and check for administrative locks before the call.

```
# Ensure the Service Principal has 'Organization.Read.All'
def revoke_licenses(user_id, access_token):
    endpoint = f"https://graph.microsoft.com/v1.0/users/{user_id}/assignLicense"
    headers = {'Authorization': f'Bearer {access_token}',
               'Content-Type': 'application/json'}

    # Correct payload for revoking all licenses
    # We must fetch active licenses first to populate 'removedLicenses'
    payload = {
        "addLicenses": [],
        "removedLicenses": active_sku_ids # populated from a prior /users/{id}/licenseDetails call
    }

    response = requests.post(endpoint, headers=headers, json=payload)
    if response.status_code == 403:
        print("Error: 403. Check 'Organization.Read.All' permissions in Azure Portal.")
    return response.json()
```

DIVING INTO CHATGPT-5 · EXAMPLE B

Root cause analysis — drafting a post-mortem.

PROMPT

"I am drafting an RCA for a 2-hour outage in our Slack-to-Jira integration. Based on these raw incident logs: [Paste Logs], write a professional post-mortem. Structure it with: **Executive Summary**, **Technical Root Cause**, and **3 Preventative Actions**. Use a tone that is accountable yet confident."

Incident report — Slack ↔ Jira integration outage

April 21, 2026
Date

120 minutes
14:00 – 16:00 UTC

Resolved
Status

1 · Executive summary

The bi-directional sync between Slack and Jira experienced a total service disruption for approximately two hours. Service was restored by rolling back a faulty webhook listener configuration. No data was lost, though a backlog of 450+ events required manual reconciliation.

DIVING INTO CHATGPT-5 · EXAMPLE B

Technical root cause & preventative actions.

2 · Technical root cause

Logs (specifically **E_TIMED_OUT** at 14:04:12) indicate a **webhook deadlock**.

- **Trigger:** a minor update to the Amazon Transform middleware introduced a recursive loop — a Jira status update triggered a Slack notification, which the middleware re-interpreted as a new Jira update.
- **Result:** an exponential surge in API calls hit Slack's rate limits (**429 Too Many Requests**) and ultimately crashed the listener (**OOMKill**).

3 · Preventative actions

- **Idempotency keys.** Mandatory idempotency headers for all Amazon Transform traffic to ignore duplicate event IDs.
- **Circuit breaker.** Auto-kill the webhook sync if the error rate exceeds 15% in a 60-second window.
- **Enhanced monitoring.** Configure Amazon Q to alert on recursive traffic patterns rather than just standard uptime pings.

EXPLORING MICROSOFT COPILOT

The corporate navigator.

Copilot excels at finding information buried in internal Microsoft Graph — Teams, Outlook, SharePoint.

OPS USE CASE

Internal search

Pull SOPs, decisions, and threads out of channels you barely remember joining.

"Find the SOP for 'Firewall Exception Requests' that was shared in the 'Security-Ops' Teams channel last month."

FEATURE

Copilot Agents

Mini-AI assistants that live inside Teams to handle specific tasks — onboarding a new hire, checking ticket status, escalating outages.

INTEGRATION

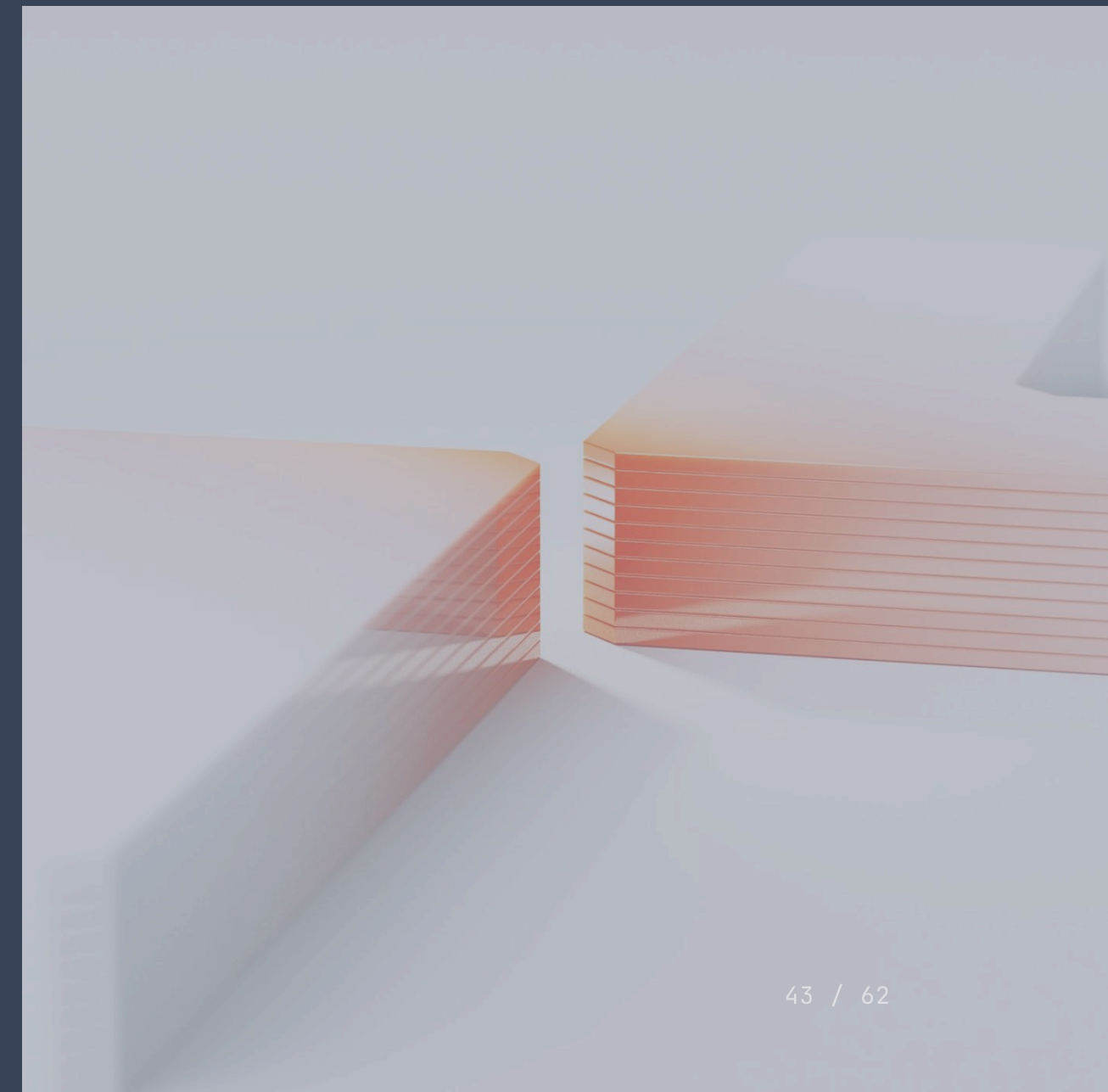
Microsoft 365

Seamless connectivity across Outlook, Teams, and SharePoint — one query against your organizational knowledge.

PART 04 • MODULE

Ethics & Governance in AI.

Drawing the line between augmentation and abdication — for our clients, our data, and ourselves.



GOVERNANCE WORKFLOW

Maintaining 100% of human accountability.

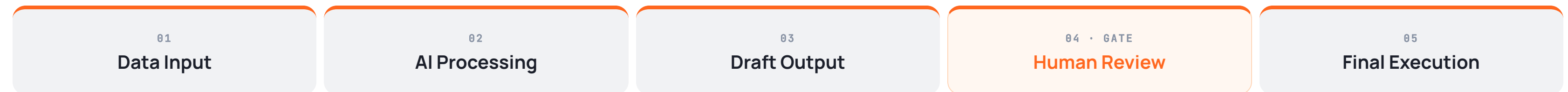
THE GOLDEN RULE

AI drafts. Humans approve.

Captures 80% of efficiency gains while maintaining 100% of accountability. Over-relying on AI responses without human review is the most common — and the most expensive — mistake.

HUMAN-IN-THE-LOOP ARCHITECTURE

Five gates between data & **final execution**.



Review checklist

- **Factual accuracy:** right product name, price, policy?
- **Tone calibration:** appropriate for this specific situation?
- **Completeness:** did it address all parts of the question?
- **Hallucinations:** did it invent features or commitments?
- **Personalization:** does it read like a robot?

When NOT to use AI

- Legal / compliance issues.
- High-stakes emotional escalations (loss, grief, anger).
- Executive communications.
- Anytime AI lacks the data it actually needs.

THE ETHICAL IMPERATIVE

Tool vs. **behavior.**

The new stakes: creators cannot always explain the output, and generation runs at massive scale — text, images, deepfakes, instantly.

The tool

Technology is entirely neutral until it is applied. We cannot simply blame the algorithm.

The behavior

How we deploy it — the guardrails we remove and the data we feed it — defines the ethical outcome.

THE MANDATE

Move from asking "*can we build this?*" to "*should we build this?*"

THE ETHICAL AI FRAMEWORK

Three pillars to **evaluate any AI initiative.**

01 • INTENT

The why

Is the primary goal to help the user, or is it to manipulate behavior?

02 • PROCESS

The how

How was the model built? Does it have data hygiene (removing sensitive identifiers) and a paper trail for auditability?

03 • OUTCOME

The what

Who wins, who loses? Does it create a digital divide? Are users notified they are interacting with AI (transparency)?

THE REAL-WORLD TOOLS LANDSCAPE

Across support & ops.

#	CATEGORY	PLATFORM
01	HELPDESK • SUPPORT	Zendesk AI • Freshdesk Freddy AI
02	CRM • BOTH	Salesforce Einstein
03	INCIDENT MGMT • OPS	PagerDuty AIOps
04	OBSERVABILITY • OPS	Datadog Watchdog • Dynatrace Davis AI
05	CONVERSATIONAL AI • SUPPORT	Intercom Fin
06	KNOWLEDGE MGMT • BOTH	Notion AI • Confluence AI
07	GENERAL AI • BOTH	Microsoft Copilot • Claude • ChatGPT

APPROVED AI DIRECTORY – CORE MODELS & CLOUD PLATFORMS

Authorized for **general knowledge work.**

Authorized foundational LLMs and enterprise infrastructure for secure deployment.

CONVERSATIONAL AI & WORKSPACES

General chat & reasoning

- ✓ **Gemini for Google Workspace**
Company-wide license — everyone can access.
- ✓ **ChatGPT-5**
Advanced reasoning and drafting capabilities.
- ✓ **Claude AI**
Large-context analysis and document summarization.

ENTERPRISE CLOUD ECOSYSTEMS

Infrastructure & API access

- ✓ **Azure OpenAI**
Secure enterprise deployment and integration.
- ✓ **Vertex AI**
Custom model building and ML operations.
- ✓ **Amazon Bedrock**
Scalable foundation-model API access.

APPROVED AI DIRECTORY – SPECIALIZED & COLLABORATIVE TOOLS

Authorized point solutions.

ENGINEERING & DEV

Code & tech ops

✓ **GitHub Copilot**
AI pair programmer.

✓ **Cursor**
AI-first code editor.

✓ **Amazon Q & Transform**
Business / dev assistant; code modernization.

DESIGN & TEAM

Assets & knowledge

✓ **Figma AI**
Design generation and asset management.

✓ **Rovo · Atlassian**
Enterprise search and knowledge graphing.

✓ **Slack**
Thread summarization & workflow automation.

PRODUCTIVITY & UTILITY

Translation & whiteboarding

✓ **DeepL**
High-accuracy neural machine translation.

⚠ **Miro**
Approved with condition – users MUST configure settings using the 'opt-out' option.

CHOOSING THE RIGHT TOOL

Four questions before you **commit**.



QUESTION 01

Where does my data go?

Does the vendor train models on your inputs?
Read the data-residency & retention terms.



QUESTION 02

Can I see why?

Is there explainability – can you trace why
the AI made a decision?



QUESTION 03

How do I correct it?

What's the feedback loop when the AI is
wrong? Who owns the retraining?



QUESTION 04

What's the override?

What does the human override look like in
practice – and how fast?

GUARDRAILS, ETHICS, AND PROFESSIONAL RESPONSIBILITY

Six risk domains.

RISK • HALLUCINATIONS

Verify against authoritative sources.

AI fabricates terms. Always verify before sending.

RISK • HUMAN-IN-THE-LOOP

Refunds, terminations, legal.

These require human judgment, full stop.

RISK • BIAS

Audit disparate impact.

Inherits historical biases (e.g., ticket routing favoring demographics).

RISK • COMPLIANCE

GDPR • CCPA • HIPAA.

All apply to AI-processed data.

RISK • DATA PRIVACY

Never paste PII or credentials.

Especially into public tools.

RISK • AUDITABILITY

Maintain decision audit trails.

If you can't explain it, you can't ship it.

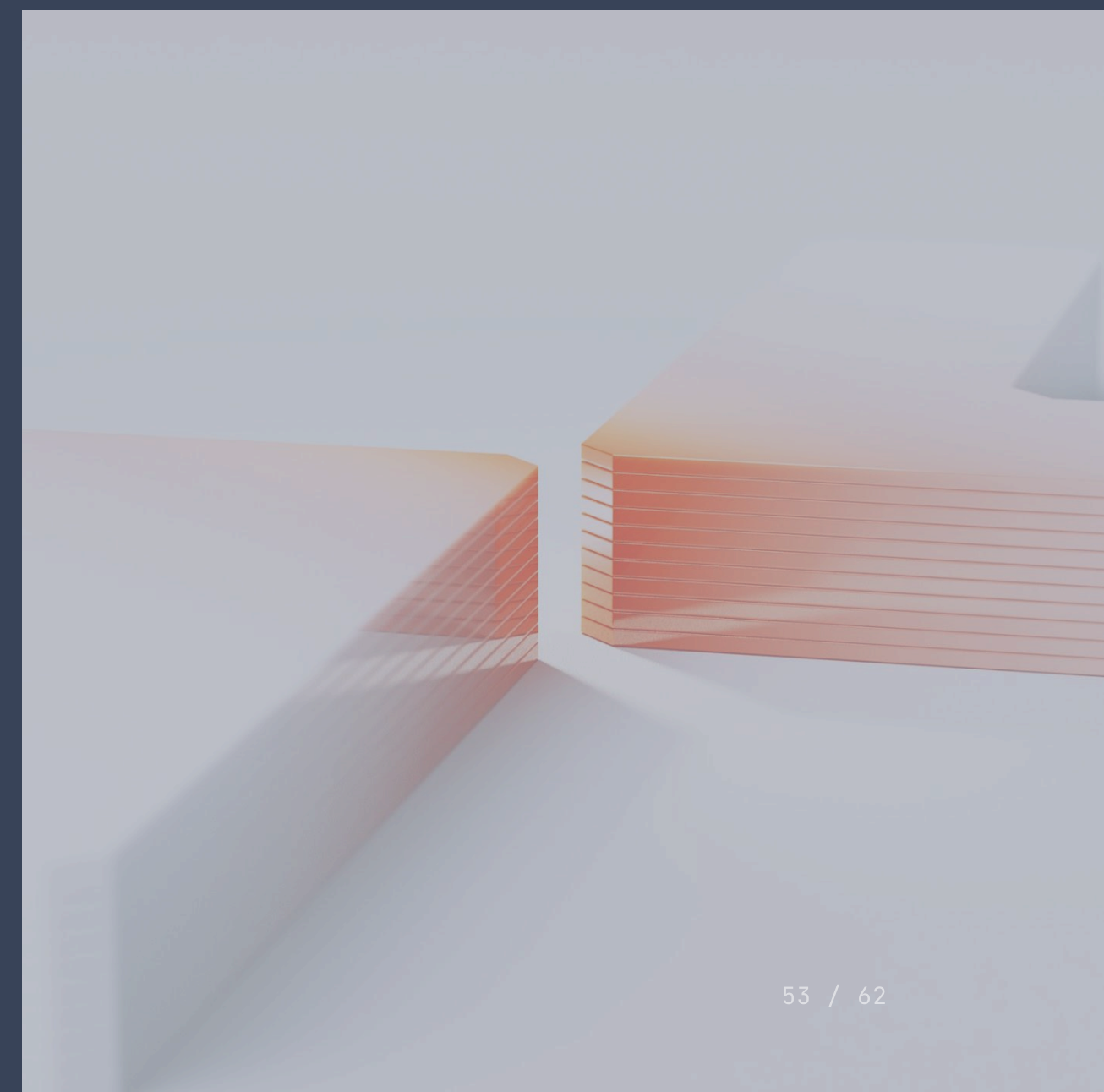
THE PRACTICAL CHECKLIST

- Know your organization's approved AI tools.
- Treat every AI output as a draft until verified.
- Know your escalation path for scenarios bypassing AI entirely.
- Never input SSNs, card numbers, or full records without IT approval.
- Report AI errors to improve the models.

PART 05 • MODULE

What's next?

From answering questions to doing complete tasks intelligently — the 2025–2026 trajectory.



THE CAPABILITY BLUEPRINT

From reactive to **autonomous**.



STAGE 01 • RAG

Retrieval-Augmented Generation

The shift: from guessing to fetching real-time, external data.

Searching company leave policies accurately.



STAGE 02 • AGENTIC AI

Autonomous task execution

The shift: from AI that talks to AI that does work.

Reading emails and booking meetings automatically.



STAGE 03 • AGENTIC RAG

Reasoning + retrieval

The shift: multi-step reasoning combining iterative searches with decision-making.

An AI research assistant combining insights from multiple sources.

SHAPING THE 2025-2026 ENTERPRISE LANDSCAPE

Seven forces.

∞

01

Agentic Workflows

End-to-end autonomous execution. Expected in 40% of enterprise apps by 2026.



02

Small Language Models

Domain-specific, faster, and highly cost-efficient.



03

Multimodal AI

Seamlessly processing text, image, audio, and video together.



04

Human + AI Collab

Evolving from a simple tool into an active digital co-worker.



05

Hyperautomation

Autonomous execution of entire business processes — supply chain decisions, etc.



06

Governance & Safety

Expanding focus on risk management, compliance, and ethics.



07

AI-native Systems

Software architected with AI at its core from the design stage.

BOTTOM LINE

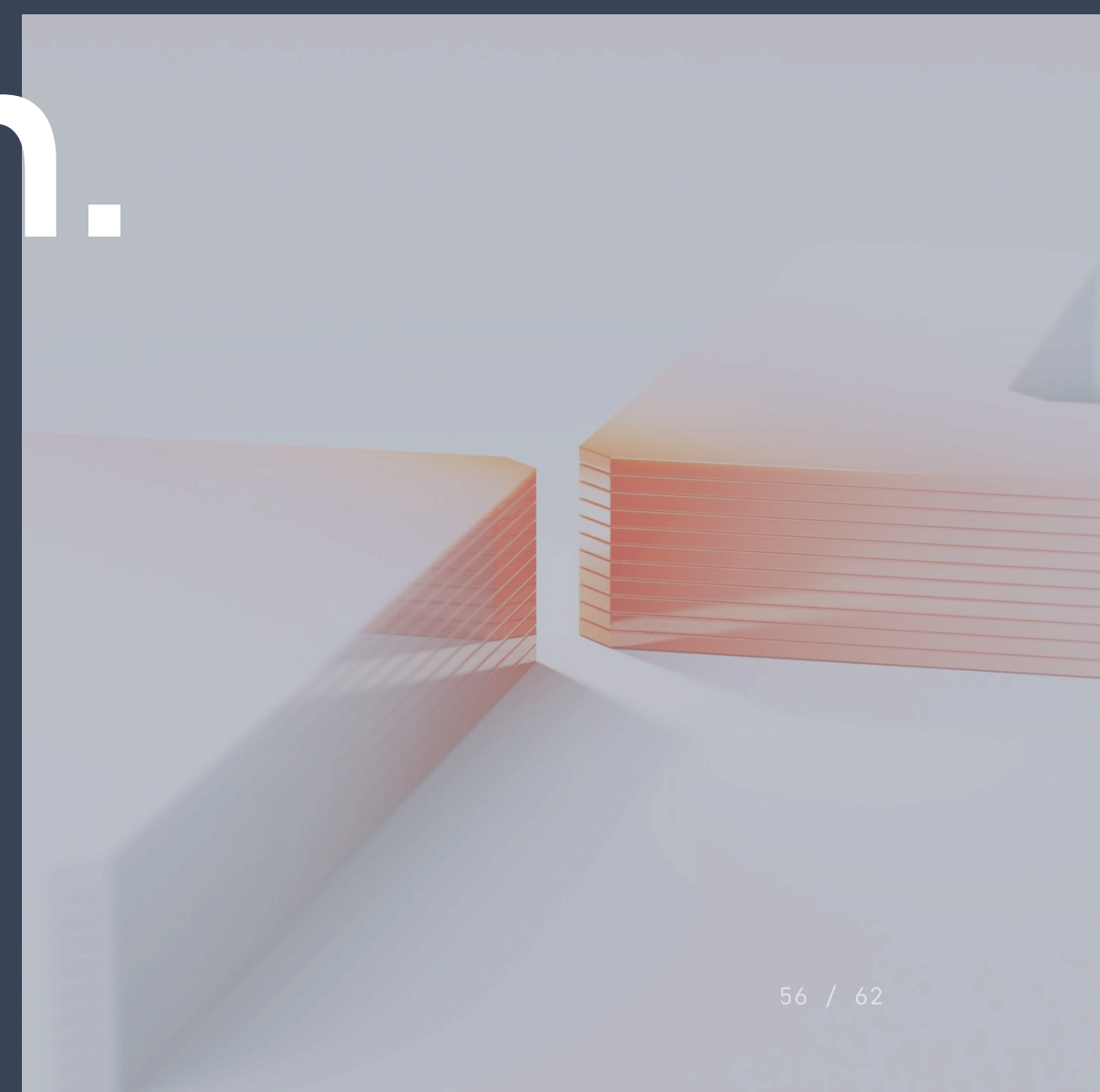
From answering to *doing*.

"AI is evolving from answering questions to doing complete tasks intelligently."

PART 06 • MODULE

Mastering daily integration.

Becoming an AI-augmented professional, not an AI-dependent one.



DRIVING ADOPTION • FROM EXECUTION TO OVERSIGHT

"Will AI take my job?" The role isn't going away — it's **shifting** from answering tickets to managing the AI that answers tickets.

The goal is AI-augmented professionals, not AI-dependent ones.

5-STEP TRUST LADDER

Build **organizational trust** in AI.

STEP 01

Start low-stakes

Summaries, notes, drafts — places where mistakes are cheap.

STEP 02

Track accuracy

As-is vs. edited. Quantify what AI gets right out of the box.

STEP 03

Celebrate wins

Time saved, tickets deflected. Make the upside visible.

STEP 04

Be transparent on failures

Treat them as learning data, not blame.

STEP 05

Give AI feedback

Thumbs up / down — close the loop.

SUPPORT METRICS

AHT · FCR · CSAT/NPS · Ticket Deflection Rate · Agent Response Time

OPS METRICS

MTTD · MTTR · Alert-to-noise ratio · SLA breach rate · Incident recurrence

THE SKILL-VALUE SHIFT MATRIX

Where **your hours** compound.

Declining in value

- Manual ticket sorting / copy-paste triage.
- Basic knowledge-article retrieval.
- Template-based first responses.
- Manual log parsing / grep debugging.
- Static threshold-based alerts.

Rising in value

- ✓ Prompt engineering & AI workflow design.
- ✓ AI output quality assurance.
- ✓ Process design for human-AI collaboration.
- ✓ AI governance and ethics judgment.
- ✓ Interpreting AI-generated analytics.

THE STRATEGIC MINDSET

Three **axioms** to operate by.

AXIOM 01

AI is a tool, not a replacement.

AI's domain — handles volume. Drafting, pattern matching, baseline generation.

Human's domain — handles nuance. Judgment, empathy, contextual understanding.

AXIOM 02

Garbage in, garbage out.

The quality of your prompt determines the quality of your output.

60 SECONDS = 10 MINUTES

Invested in a better prompt. Saved in editing.

AXIOM 03

Always verify before sending.

AI drafts → humans approve.

Capturing efficiency. Maintaining accountability.

THE STRATEGIC MINDSET · LONG-TERM IMPACT

Two more **non-negotiables.**

AXIOM 04

Understand AI's failure modes.

Hallucinations

Data Privacy

Bias

Knowing these risks makes you a safer and more effective AI user — not a paranoid one.

AXIOM 05

Learn AI to outperform.

A widening competitive gap separates AI-fluent professionals from AI-avoidant ones — quarter by quarter.

Start building the skill now. Today, not next quarter.

Thank you.

L&D CATALOG

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COURSE

AI · 104 — Support & Operations Professionals

AUDIENCE

GlobalLogic engineering teams